GAP-10I: Flat Representers and the Connection-Representation Obstruction

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Abstract

The UBT tetrad is constrained by $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$, so the local rank theorem alone does not establish that a single Θ generates the required tetrads. This note closes two sharply defined subproblems. First, every constant Lorentz tetrad, and in particular Minkowski spacetime, has an explicit affine Θ representer in the flat inertial sector; it also solves the standard source-free second-order master operator because all second derivatives vanish. Second, a generic invertible Θ with a naive one-sided connection cannot support nonzero curvature under torsion-free tetrad compatibility. The natural biquaternionic two-sided/bimodule derivative avoids this flatness obstruction: its left and right curvatures need only be conjugate through Θ. This does not solve curved-space integrability, but it rigorously narrows the admissible canonical connection structure.

1 Setup

Let $\mathbb{B} = \mathbb{C} \otimes \mathbb{H}$ and let the Lorentz slice be

$$W_L = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\}. \tag{1}$$

For a UBT field Θ define

$$E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta, \qquad \frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1}. \tag{2}$$

The issue addressed here is existence and compatibility of Θ, not the already proved rank of ${e_\mu}^a \mapsto g_{\mu\nu}$.

2 Exact flat representer

Theorem 2.1 (Affine Θ representer for a constant tetrad)**.** *Let $\bar{E}_\mu \in W_L$ be four constant, linearly independent biquaternions. In a flat inertial gauge with $\Omega_\mu = 0$, define*

$$\boxed{\Theta_{\text{aff}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\,\bar{E}_\mu x^\mu,} \tag{3}$$

where $\Theta_0 \in \mathbb{B}$ is constant. Then

$$\mathcal{N}_0^{-1/2}\partial_\mu\Theta_{\text{aff}} = \bar{E}_\mu, \tag{4}$$

and Eq. (2) yields the constant Lorentz metric

$$\bar{g}_{\mu\nu}\mathbf{1} = \frac{1}{2}(\bar{E}_\mu^\sharp\bar{E}_\nu + \bar{E}_\nu^\sharp\bar{E}_\mu). \tag{5}$$

Proof. Differentiating Eq. (3) gives $\partial_\mu\Theta_{\text{aff}} = \sqrt{\mathcal{N}_0}\bar{E}_\mu$. The metric statement is the central anticommutator identity on W_L. □

Corollary 2.2 (Minkowski UBT vacuum representer)**.** *Choose*

$$\bar{E}_0 = i\mathbf{1}, \qquad \bar{E}_k = \mathbf{e}_k. \tag{6}$$

Then

$$\boxed{\Theta_{\text{SR}}(x) = \Theta_0 + \sqrt{\mathcal{N}_0}\left(ix^0\mathbf{1} + x^k\mathbf{e}_k\right)} \tag{7}$$

induces $g_{\mu\nu} = \operatorname{diag}(-1, 1, 1, 1)$. All second spacetime derivatives vanish. Therefore any flat source-free master operator whose principal part is the constant-coefficient covariant Laplacian or d'Alembertian annihilates Θ_{SR}.

Remark 2.3. This is stronger than merely saying that one may set $\Omega_\mu = 0$ in special relativity: it gives an explicit single-field UBT configuration whose first derivatives are the Minkowski tetrad. It does not by itself fix boundary conditions in compact imaginary time or prove uniqueness of the vacuum.

3 Why a naive one-sided connection is too restrictive

Consider a one-sided connection on Θ,

$$D_\mu^L\Theta = \partial_\mu\Theta + A_\mu\Theta, \qquad F_{\mu\nu}^L = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{8}$$

Then

$$[D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L\Theta. \tag{9}$$

Theorem 3.1 (One-sided invertible-field flatness obstruction)**.** *Assume on an open set that*

1. *$E_\nu = \mathcal{N}_0^{-1/2}D_\nu^L\Theta$;*
2. *the generated tetrad obeys torsion-free antisymmetric compatibility, $\mathscr{D}_{[\mu}E_{\nu]} = 0$;*
3. *the same one-sided connection is used in the induced differentiation of $D_\nu\Theta$;*
4. *Θ is invertible as an element of $\mathbb{B} \simeq \operatorname{Mat}(2, \mathbb{C})$.*

Then $F_{\mu\nu}^L = 0$ on that open set.

Proof. Because the affine connection is torsion free, antisymmetrising the covariant Hessian of the spacetime-scalar/internal field Θ gives

$$\sqrt{\mathcal{N}_0}\,\mathscr{D}_{[\mu}E_{\nu]} = [D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L\Theta. \tag{10}$$

The left-hand side vanishes by assumption, hence $F_{\mu\nu}^L\Theta = 0$. Right multiplication by Θ^{-1} gives $F_{\mu\nu}^L = 0$. □

Corollary 3.2. *A generic curved classical UBT branch cannot use the naive one-sided regular representation together with invertible* Θ *and torsion-free tetrad compatibility. Non-invertible stabilizer solutions may evade the theorem but are nongeneric and cannot be assumed to represent arbitrary GR geometries.*

This theorem closes a route rather than the whole gap: the one-sided derivative is not an acceptable generic curved-GR mechanism under the stated assumptions.

4 The natural two-sided biquaternionic derivative

Biquaternions form a bimodule over themselves, so left and right multiplication are equally intrinsic. Consider

$$\boxed{D_\mu \Theta = \partial_\mu \Theta + A_\mu \Theta - \Theta B_\mu,} \tag{11}$$

with

$$F^A_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \tag{12}$$
$$F^B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu]. \tag{13}$$

The relative minus sign is conventional and makes the right connection transform in the standard bimodule form.

Theorem 4.1 (Two-sided curvature identity)**.** *For Eq.* (11)*,*

$$\boxed{[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}.} \tag{14}$$

Proof. Expand $D_\mu D_\nu \Theta - D_\nu D_\mu \Theta$. Mixed terms containing one ordinary derivative and one connection cancel pairwise, as do the terms $A_\mu \Theta B_\nu$ after antisymmetrisation. The remaining left products form $F^A_{\mu\nu}\Theta$ and the remaining right products form $-\Theta F^B_{\mu\nu}$. □

Corollary 4.2 (Nonzero curvature is compatible with invertible Θ)**.** *Under torsion-free antisymmetric tetrad compatibility, the integrability condition is*

$$F^A_{\mu\nu}\Theta = \Theta F^B_{\mu\nu}. \tag{15}$$

If Θ *is invertible, this becomes*

$$\boxed{F^A_{\mu\nu} = \Theta F^B_{\mu\nu}\Theta^{-1}.} \tag{16}$$

Unlike the one-sided case, neither curvature must vanish. The field Θ *acts as an intertwiner between the left and right curvature representations.*

Remark 4.3 (What has and has not been selected). The theorem alone does not derive A_μ, B_μ, their relation under the UBT involutions, or their action from the canonical action. It proves a structural constraint and exhibits a minimal algebra-native escape from the one-sided obstruction: a general two-sided/bimodule connection can retain invertible Θ, torsion-free compatibility, and nonzero curvature simultaneously.

Remark 4.4 (Subsequent pure-pair audit). Lorentz-slice and metric compatibility select the no-new-field representative $A_\mu = \Omega_\mu$, $B_\mu = -\Omega^\ddagger_\mu$ modulo a cancelling central term. With $K = 0$ that representative is closed as a concurrent-vector no-go for the non-flat Schwarzschild vacuum exterior with nonzero mass. The no-go is torsion-free. A companion theorem constructs an explicit local single-Θ representer for every smooth tetrad using composite metric-compatible contortion inside the same one-connection pair. A relative bimodule action is therefore optional for local kinematics and remains a possible torsion-free composite/auxiliary route. See `canonical/gr_closure/gap_10i_paired_connection_audit.tex` and `canonical/gr_closure/gap_10i_torsionful_local_representer.tex`.

5 Implicit and possibly transcendental self-consistency

Once the classical connection is reconstructed from the tetrad and torsion, Eq. (11) becomes schematically

$$E_\mu = \mathcal{N}_0^{-1/2} \left[\partial_\mu \Theta + A_\mu[E, T]\Theta - \Theta B_\mu[E, T] \right]. \tag{17}$$

This is an implicit nonlinear first-order PDE or fixed-point system because E appears on both sides through the connection. It is not automatically a transcendental equation in the strict mathematical sense. However, when $\Theta(q, \tau)$ is restricted to a Jacobi-theta or other transcendental function class, the complete UBT system may be both implicit and transcendental. These are distinct properties and should be stated separately in formal proofs.

6 Revised partial closure of GAP-10I

Defensible status

GAP-10I-SR: CLOSED [L1]. Every constant Lorentz tetrad has the explicit affine representer Eq. (3); Minkowski spacetime is therefore generated by a single Θ in the flat inertial branch.

GAP-10I-1S: CLOSED AS NO-GO [L1]. A naive one-sided regular connection with invertible Θ cannot support generic nonzero curvature under torsion-free tetrad compatibility.

GAP-10I-PAIR-KIN: CLOSED [L1]. The pure Lorentz-compatible pair reduces to one spin connection modulo a central term that cancels.

GAP-10I-PAIR-GR: CLOSED AS A TORSION-FREE NO-GO [L1]. With $K = 0$ the pure pair implies a concurrent vector and excludes the non-flat Schwarzschild vacuum exterior with nonzero mass.

GAP-10I-TORSION-LOCAL: CLOSED LOCALLY [L1]. The same one-connection pair with explicit composite metric-compatible contortion has a local single-Θ representer for every smooth Lorentzian tetrad.

GAP-10I-2S: NOT REQUIRED FOR LOCAL KINEMATICS. A relative bimodule action remains an optional torsion-free completion and, if used, must be derived as composite or auxiliary and shown nonpropagating.

GAP-10I-PRESCRIBED: CLOSED [L1]. For specified smooth (E, A, B) the exact solution and path-independence problem is the augmented holonomy criterion proved in `gap_10i_augmented_holonomy.tex`.

GAP-10I-CURVED: LOCAL KINEMATICS CLOSED; DYNAMICS/-GLOBAL PART NARROWED. Local representability is explicit. Canonical action-level selection, physical torsion constraints, regularity, and global continuation for Schwarzschild, Kerr, FRW, and generic Einstein geometries remain to be proved.